

PYTHON +ML+DL

1. PYTHON INTRODUCTION: 15 HOURS

- History • Features • Why Python for ML?: 0.5h
- Basic Python Programming (Syntax, Variables, Operators, Conditionals, Loops): 2h
- String Manipulation (Accessing, Operations, Slices): 1h
- Lists (Intro, Access, Operations, Functions): 1h
- Tuples (Intro, Access, Operations, Functions): 1h
- Dictionaries (Intro, Access, Working, Properties): 1h
- Functions (Defining, Arguments, Anonymous, Global/Local): 1.5h
- Modules (Import, Math, Random, Packages): 1h
- Exception Handling (Except, Try-Finally, User Defined): 1h
- OOPs (Class/Object/Attributes, Inheritance, Overloading): 5h

2. PYTHON LIBRARIES: 12 HOURS

- NumPy: Installing, Arrays (1D/2D/Multi), Random, Indexing/Slicing, Shape/Reshape, Join/Split/Search/Sort/Filter: 5h
- Pandas: Intro/Setup, Data structures, Loading, Indexing/Selection, Text/Aggregation, Missing Data, Grouping/Merging, Date/Multi-Index: 5h
- Matplotlib: Plotting, Subplots, Markers, Line style, Charts/Graphs: 1.5h
- Seaborn: Multiple charts: 0.5h

3. BASIC STATISTICS: 4 HOURS

- Mean • Mode • Median • Variables • Standard Deviation • Outliers • Hypothesis: 4h

4. SQL: 8 HOURS

- Database basics, Types, DBMS vs RDBMS, Normalization/Denormalization: 1h
- MySQL Install, Data Models, DBMS Language, ACID: 1h

- SQL: Data Types, Commands, Operators, Keys, Joins: 2h
- GROUP BY/HAVING/ORDER BY, Subqueries (SELECT/INSERT/UPDATE/DELETE): 1.5h
- Set Operations, Functions, SQL vs NoSQL, Python Connection: 2.5h

5. DATA PREPROCESSING: 5 HOURS

- Overview, Get Database, Import Libraries/Datasets: 1h
- Missing data, Categorical Data: 1h
- Splitting (Train/Test), Feature Scaling: 1.5h
- Working directories: 0.5h

6. INTRODUCTION OF ALL: 2 HOURS

- Machine Learning • Deep Learning • AI • Data Science: 2h

7. MACHINE LEARNING: 25 HOURS

- 7.1 Regression: Linear Theory + Hands-on (House prices), Polynomial/Overfitting, RMSE/R², Regularization + Project: 6h
- 7.1 Classification: Logistic + Diabetes, Metrics (Precision/Recall/F1/ROC), Trees/RF/SVM + Spam Project: 6h
- 7.2 Ensemble: Intro, Max Voting/Averaging/Weighted, Stacking/Blending/Bagging/Boosting, AdaBoost/XGB: 4h
- 7.3 Performance: Confusion Matrix, Accuracy, Report (Precision/Recall/F1), R2Score: 2h
- 7.4 Unsupervised: K-Means + Customers, Hierarchical + Movies, PCA + Project (Customer segments): 7h

8. ML CAPSTONE: 8 HOURS

- 5 Projects assignment: 8h

9. NEURAL NETWORKS + DL BASICS: 10 HOURS

- Intro Neural Networks + TensorFlow/Keras hands-on: 2h
- CNNs + MNIST digits: 2.5h
- RNNs + Text generator: 2h
- Transfer Learning + Image classification project: 3.5h

10. ADVANCED + CAPSTONE: 11 HOURS

- NLP Intro + Sentiment analysis: 2h
- Time Series + ARIMA/LSTM sales forecast: 2.5h
- Reinforcement Learning + Game simulation: 1.5h
- DL Capstone: Recommendation/Medical/Chatbot project: 5h